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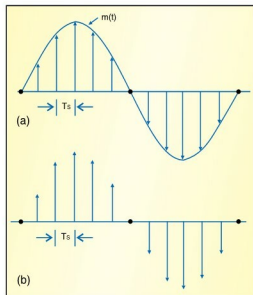


Fig. 2: The process of sampling of a sine-wave signal sampled at the rate T_s

signals in digital communication?

Shimbu

Through e-mail

A2. Analogue signals are formatted or coded into digital signals to make them compatible with digital processing. While there are many ways of implementing the processing, the three conceptual steps are sampling, quantisation and coding.

Sampling. An analogue signal is continuous in time. The signal has a level at every instant. If we want to digitise every instantaneous value of the analogue signal, the storage/processing medium required will be enormous. Here, the sampling theorem by Shannon comes to our rescue. The theorem states that if a signal $m(t)$ is band-limited and real-valued, it can be uniquely determined from its values $m(nT_s)$ sampled at uniform intervals $T_s \leq 1/(2f_m)$, where f_m is the highest frequency in the band-limited signal.

In other words, it is unnecessary to convert every instantaneous value into a digital signal. It is sufficient if a minimum number of samples of the analogue signal are converted into digital signal. The question is what should be the minimum sampling rate. As stated above, the minimum sampling rate is $2f_m$ samples per second (also

known as Nyquist rate). Generally, sampling is done using a sample-and-hold circuit. The process of sampling of a sine-wave signal sampled at the rate T_s is shown in Fig. 2.

Quantisation. While sampling converts a time signal into a discrete time signal, quantisation converts a continuous amplitude sample into a discrete amplitude sample. In other words, continuous voltage signals are mapped to a discrete (quantised) number of voltage levels. Thus sampling and quantising an analogue signal yield a digital signal. Since digital computers are binary in nature, the number of quantisation levels is usually a power of '2', i.e.,

$$N = 2^n$$

where 'n' is the number of quantisation bits.

Coding. Coding is the process of converting the quantised signals into a digital representation. It is performed by giving each quantisation level a unique label. For instance, if four bits are used, the lowest level may be 0000 (in binary), the next highest level 0001, and so on.

Quantisation and encoding operations are usually performed in the same analogue-to-digital converter circuit.

In short, sampling is the process in which an analogue signal is sampled by measuring its amplitude at discrete instants. Representing the sampled values by a finite set of levels is called 'quantisation.' Expressing each quantised level by a code is called 'coding.'

For reconstruction of the original signal (at the receiver) from the digitally coded signal, the process of digital-to-analogue conversion and low-pass filters are used.

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